Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *WXIMPACTBENCH* | ***Context:*** *Gran Chaco Tri-Border Emergency Response Zone*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Benchmark inputs are English-language OCR-corrected historical newspaper prose averaging 2,987 tokens (long-context) or 781 tokens (mixed-context) [Q109]. Models must support 8k token input lengths [Q55] and handle 'outdated terminology, spelling variations, and evolving writing conventions' [Q14]. Deployment requires processing informal Rio Platense Spanish social media (typically 20–280 tokens per post) with platform abbreviations, hashtags, indigenous-language toponyms, and code-switching [WEB-11, regional YAML register_variation]. The noise patterns in the benchmark (OCR column-fragmentation artifacts [D11, D17, D21]) are unrelated to the noise patterns in the deployment (informal orthography, abbreviations like 'tmbn', 'q pasa'). RoBERTuito-class social-media-tuned Spanish models exist [WEB-11] but the benchmark provides no evaluation in that direction.

ID	Evidence
Q14	“Historical newspapers often employed more descriptive and elaborate narratives compared to modern reporting styles (Bingham, 2010). These narratives frequently included outdated terminology, spelling variations, and evolving writing conventions (Campbell, 2013).” (p.2)
Q55	“The used models for the two tasks cover different sizes and support the input length of at least 8k tokens.” (p.6)
Q109	“The average number of tokens per article is 2987.4 in long-context settings and 781.3 in mixed-context settings.” (p.15)
D11	-2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25 JS 1 Almaden 1000 35 35 35 — Stock ticker/financial table — all-zero label, clearly off-topic content
D17	01 SECOND RACE: Trot, I Mile Purse: 11,000 5 Prime Saliora (M. Lalonde) 9.70 5.10 5.70 3 Bidou Fio (R. SimarrJ) 5.30 4.40 — Horse racing results table with unrelated editorial on optometrists' ad — off-topic noise
D21	TRAFFIC IS PARALYZED In Western Canadian Cities, and at Many Points In the United States Disasters In England. — 1894 blizzard chunk (segmented from Ex. 3) — infrastructure label carried over to chunk
WEB-11	RoBERTuito (500M+ Spanish tweets) is the most relevant existing model for the deployment's input register, but is not disaster-domain-specific and not regionally evaluated (https://aclanthology.org/2022.lrec-1.785/)

Strengths

- Mixed-context version with ~250-token chunks [Q46] is closer to deployment token lengths than long-context version
- Benchmark explicitly acknowledges register/style variation matters [Q14, Q32], even though only English historical→modern variation is tested
- Models evaluated must support 8k+ tokens [Q55], which is more than sufficient for short Spanish social media posts

ID	Evidence
Q14	“Historical newspapers often employed more descriptive and elaborate narratives compared to modern reporting styles (Bingham, 2010). These narratives frequently included outdated terminology, spelling variations, and evolving writing conventions (Campbell, 2013).” (p.2)
Q32	“Different from them, our constructed samples require the models to understand the linguistic shifts between historical and modern texts and address inconsistent styles of narratives across various periods.” (p.5)
Q46	“For the long-context impact evaluation, we create an alternate version (mixed context), whose sample length is split into segments with approximately 250 tokens from the original one (long-context version) following (Levy et al., 2024).” (p.6)
Q55	“The used models for the two tasks cover different sizes and support the input length of at least 8k tokens.” (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Significant signal distribution mismatch: English formal newspaper prose vs. Spanish informal social media; OCR-noise vs. social-media-noise; 781–2,987 token articles vs. 20–280 token posts [Q109, Q120, regional register_variation].

IF-2: Deployment infrastructure does not produce OCR-style inputs; it ingests live Twitter/X, Facebook, WhatsApp text streams [WEB-5, regional cultural_norms_notes]. Connectivity in the Gran Chaco is documented as patchy [WEB-5], which affects input freshness but not input form per se.

IF-3: Domain-specific form differences are severe: benchmark cannot evaluate handling of hashtags, platform abbreviations, voseo, indigenous toponyms, or Jopara code-mixing [WEB-9, WEB-10].

IF-4: Critical external validity violation: model performance on benchmark's English long-form OCR-corrected prose cannot predict performance on Spanish short-form informal social media posts.

ID	Evidence
Q14	"Historical newspapers often employed more descriptive and elaborate narratives compared to modern reporting styles (Bingham, 2010). These narratives frequently included outdated terminology, spelling variations, and evolving writing conventions (Campbell, 2013)." (p.2)
Q32	"Different from them, our constructed samples require the models to understand the linguistic shifts between historical and modern texts and address inconsistent styles of narratives across various periods." (p.5)
Q109	"The average number of tokens per article is 2987.4 in long-context settings and 781.3 in mixed-context settings." (p.15)
Q120	"Figure 6 presents the token length distribution of passages in two versions of our dataset: (a) the Long Context dataset and (b) the Mixed Context dataset used for context-denoising evaluation." (p.16)
D3	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed — 1894 blizzard article: infrastructure disruption to Canadian railway/electric car — exemplary on-target sample
D11	-2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25 JS 1 Almaden 1000 35 35 35 — Stock ticker/financial table — all-zero label, clearly off-topic content
D21	TRAFFIC IS PARALYZED In Western Canadian Cities, and at Many Points In the United States Disasters In England. — 1894 blizzard chunk (segmented from Ex. 3) — infrastructure label carried over to chunk
WEB-5	Cross-border displacement and indigenous community evacuation are primary Gran Chaco disaster consequences (10,000+ Wichí/Toba/Chorote/Tapiete evacuated in 2018) (https://www.iadb.org/en/staying-ahead-gran-chacos-floods)
WEB-9	No NLP tools handle indigenous-language lexical borrowing within Spanish disaster text (https://arxiv.org/html/2501.09943v2)
WEB-10	No mixed-language disaster classification tools exist for Guaraní/Quechua/Wichí lexical borrowing within Spanish (https://arxiv.org/html/2404.05365v1)
WEB-11	RoBERTuito (500M+ Spanish tweets) is the most relevant existing model for the deployment's input register, but is not disaster-domain-specific and not regionally evaluated (https://aclanthology.org/2022.lrec-1.785/)

Information Gaps

- Empirical token-length and noise-pattern distribution for Gran Chaco disaster social media has not been characterized

Requires Expert Verification

- A small Spanish-social-media replication study (e.g., evaluating RoBERTuito on Gran Chaco posts) would directly establish form-mismatch impact

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Remediation

Gap 1: Benchmark form is English long-form OCR-corrected prose; deployment form is short Spanish social media with informal orthography and code-mixing.

Recommendation: Pilot RoBERTuito and multilingual LLMs on a Spanish social-media disaster classification task, using token-length and noise-pattern distributions matching deployment inputs; report results separately from wximpactbench scores.